



Tshwane University of Technology

We empower people

SURNAME & INITIALS: MANALA LL

STUDENT NUMBER: 230325898

COURSE: COMPUTER SYSTEMS ENGINEERING

SUBJECT: PD135D PCB Report

DATE: 15 May 2026

TABLE OF CONTENTS

1. Introduction.
 - a. Summary.
2. Design.
 - a. Proteus circuit diagram.
 - b. PCB layout from Proteus.
 - c. 3D Model.
3. Mechanical Design.
 - a. Software specifications.
 - b. hardware specifications.
4. Production.
 - a. List of tools used in design and their usage
 - b. Chemical preparation.
 - c. Finished board.
5. Conclusion.
 - a. Conclusion based on task, challenges, and proposed solutions.
 - b. Recommendations.
6. Appendices.
 - a. Bill of materials.

INTRODUCTION.

This project aimed to design and fabricate a toggle switch circuit using the 555 timer IC configured in astable mode and the dual D-type flip-flop. The 555 timer generates a continuous square wave that serves as the clock input for the flip-flop, creating toggle-switch functionality. The design was simulated in Proteus 8, and the physical PCB was produced via chemical etching on a copper board with the circuit printed on it.

DESIGN

- CIRCUIT DIAGRAM FROM PROTEUS.

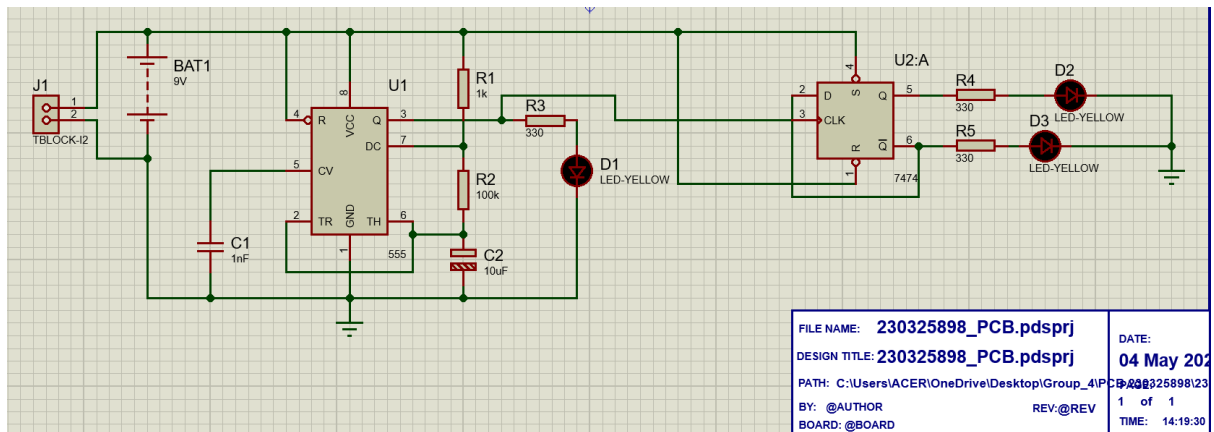


Figure 1: Proteus circuit schematic

- PCB LAYOUT FROM PROTEUS.

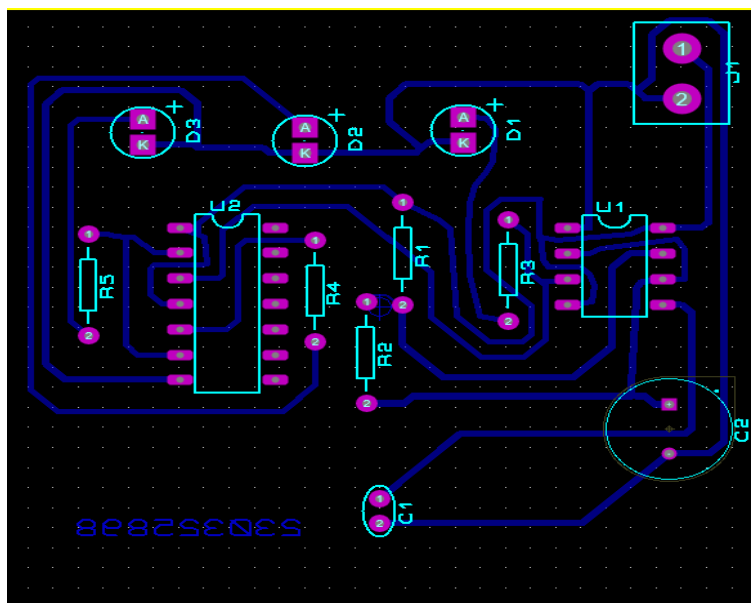


Figure 2: PCB Layout

- 3D MODEL OF THE DESIGN.

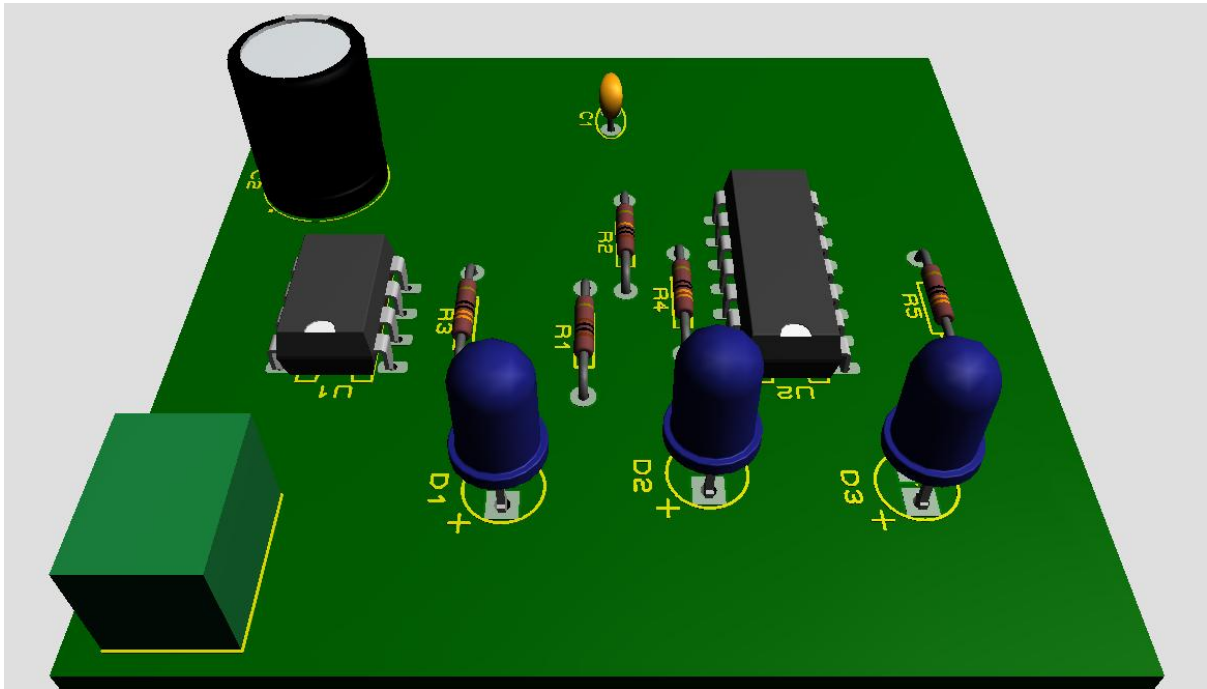


Figure 3: 3D view

MECHANICAL DESIGN.

- Software Specifications.
 - Software – Proteus 8 Professional.
 - Board shape – Square/Rectangle.
 - Board size – 75mm x 100mm.
 - Board layout – Single-layered copper (Bottom copper).
 - Board thickness – Default.
- Hardware specification.
 - Copper board – 150mm x 300mm.
 - Glossy paper.
 - Hydrochloric acid.
 - Hydrogen peroxide.
 - Drill and drill bits.
 - Solder station.

PRODUCTION

- List of tools used and their usage
 - Proteus 8 Professional – circuit simulation, schematic design, PCB Layout, and 3D visualization.
 - Printer – printing the mirrored PCB layout onto the glossy paper for toner transfer.
 - Clothes iron – transferring toner from glossy paper to copper board.
 - Hand drill – Creating holes for through-hole components.
 - Scrub/steel wool – cleaning and preparing the copper board before transfer.
 - Soldering iron – component soldering and trace repair.

- Multimeter – continuity testing and voltage verification.
- Permanent marker – Trace repair.
- Etching solution – used to etch the copper board, leaving the circuit design only.
- Thinners – Remove excess toner and marker ink.
- Containers – used to hold the etching solution, thinners, and water.
- Chemical preparation.
 - Preparation procedure.
 - Wear protective gloves and safety glasses.
 - Add warm water to a plastic container.
 - Add the hydrochloric acid to another plastic container.
 - Slowly add the hydrogen peroxide to the hydrogen peroxide.
 - A 1:2 ratio for the hydrogen peroxide and hydrochloric acid, respectively.
 - Wait for the solution to cool to room temperature.
 - Store in a sealed container away from metal surfaces.
 - Etching process.
 - Clean the copper board thoroughly with a sponge scrub or steel wool to remove grease and oxidation.
 - Print the mirrored PCB layout onto the glossy paper using the printer.
 - Align the glossy paper with the copper board and secure with tape.
 - Apply the hot iron (no steam/water) with firm pressure for 10-15 minutes.
 - Wait for the board to cool down, then place it into water.
 - Gently peel the paper, leaving the toner tracks.
 - Inspect for broken tracks and repair with a permanent marker.
 - Place the board into the etching solution, shake gently for 5-10 minutes.
 - Remove the board when unwanted copper is dissolved.
 - Rinse the board in water.
 - Remove toner with thinners.
 - Safety precaution.
 - Work in a well-ventilated area.
 - Avoid skin contact with the etching solution.
 - Wear eye, hand, nose, and mouth protection throughout the process.
- **FINISHED BOARD.**

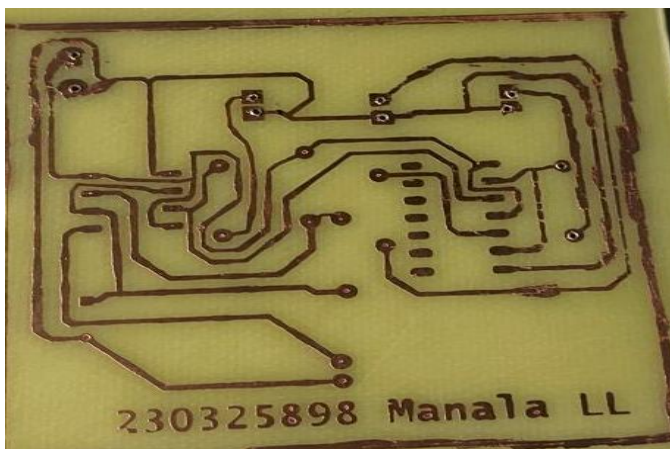


Figure 4: Finished etched and drilled PCB

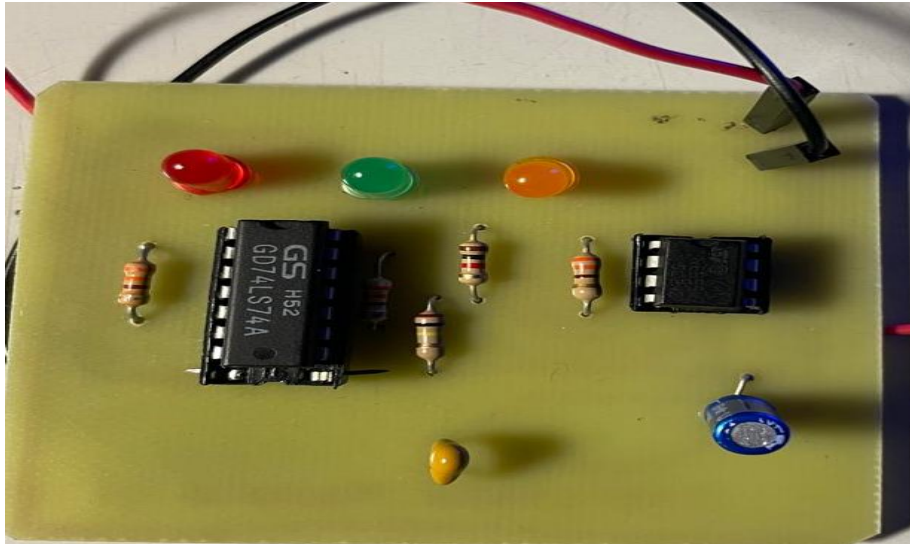


Figure 5: Fully assembled PCB

CONCLUSION.

The design and fabrication of the toggle circuit was successfully completed using Proteus 8 Professional for simulation and PCB layout, followed by physical production via the toner transfer method. Several challenges were faced during production, including

- Toner transfer not complete – increasing ironing time and pressure helps.
- LED not toggling – Verified D input connection.
- Solder joint issue – resoldered joints that were not properly soldered.

Based on these experiences, it is recommended that future iterations consider transitioning to a double-sided PCB with a dedicated ground plane for improved noise immunity, incorporating surface-mount components for a more compact design, replacing fixed timing resistors with a potentiometer to allow adjustable toggle frequency, adding a separate power-on LED for clearer status indication, and designing a 3D-printed enclosure for professional presentation and physical protection of the circuit board.

• BILL OF MATERIALS.

item	component	Part number	quantity	Unit price	Total price
1	Timer IC	NE555(DIP-8)	1	R12.50	R12.50
2	Flip-flip IC	74LS74(DIP-14)	1	R15.00	R15.00
2	Resistor	1K,330,100K	(1 x 5)	R0.50	R2.50
4	Capacitor	10uF, 1nF	(1 x 2)	R2.00	R4.00
5	LED	5mm red	(1 x 3)	R1.50	R4.50
6	Copper board	15mm x 30mm	1	R62.75	R62.75
7	Glossy paper	A4	1	R20.00	R20.00
8				TOTAL	R121.25